

Jason Mathew

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Sciences, Philadelphia, PA
Robotics, M.S. | Mechanical Engineering, B.S. — Minors: *Computer Science, Math*

May 2027
GPA: 3.92/4.0

RELEVANT EXPERIENCE

Zoox | *Technical Program Management, Hardware Development*

May 2026 – Present | San Francisco, CA

- Launching first standardized, end-to-end development process for future autonomous vehicles, creating reusable framework for 3+ programs; map deliverables and dependencies across 40+ cross-functional teams from initial concept to production
- Architecting a centralized cost-request system to intake, route, and track 15+ weekly analyses. Targeting 30% faster turnaround through automated prioritization and routing; integrating AI validation features with current financial systems

SpaceX | *Hardware Product Design, Falcon Avionics Core Engineer*

May 2025 – August 2025 | Los Angeles, CA

- Redesigned common engine-controller enclosure (9 per vehicle) to be compatible with multiple PCB revisions; use Ansys to validate structural durability under 40x flights; increased build-production output 8% by removing unreliable features
- Predicted reliability of high-risk PCBAs using Ansys Sherlock to assess PCB component response to fatigue, shock, and vibe
- Lead critical booster component qualification to pass vibration and thermal fatigue; achieved buyoff in < 6 weeks
- Investigated hardware reliability issue of tank RTDs/cameras; root cause analysis, CT scans, and Python data analysis
- Designed \$30k cheaper, in-house dyno braking system to test high-voltage motors; accelerated project timeline by 1 month

Penn Electric Racing (FSAE) | *Mechanical Lead*

May 2025 – Present | Philadelphia, PA

- Technical Program Manager of 60+ engineers in 6 mechanical systems: drivetrain, battery, chassis, aerodynamics, suspension, vehicle dynamics. Make vehicle-level architecture design decisions for performance and reliability
- Design critical components across vehicle; cell module enclosure, steering system, drivetrain mounting, chassis tabs
- Manage electromechanical integration for powertrain modules, tractive system, wiring harnesses, and 30+ sensors
- Finished entire build 2 months faster than previous years with 80% more on-track testing and 150+ miles driven
- 40% faster fabrication time for complex assemblies from coordinating GD&T, integration, and DFM/DFA reviews in CAD
- Driver Systems Lead
 - Placed top 10 in Design Event at competition with first principles-based design. See [Design Brief Slide Deck](#)
 - Owned design of pedals, brakes, steering, dashboard, ergo and managed cross-subsystem integration: performance tradeoffs, structural and thermal FEA simulations, CNC and plastic manufacturing for 40+ parts, testing validation

Precision Machining Lab Assistant | *Teaching Assistant*

August 2025 – Present | Philadelphia, PA

JHU APL, Force Projection Sector | *Autonomous Systems Design Engineer*

September 2022 – August 2023 | Laurel, MD

- Secret Level Security Clearance, working on 3+ software projects for Guidance, Navigation, and Controls team
- Developed multiparameter design optimization model for aerodynamic systems in MATLAB & C++, reduced runtime 60%
- Contributed to MATLAB aerodynamic model for hypersonic airplanes, meshing and using panel methods to estimate C_p

PROJECTS

HMS Switch Interface | *Circuit Architect, Penn ADAPT*

January 2024 – May 2024

- Designed, routed, and soldered PCBs for switch interfaces—assistive technology for individuals with cerebral palsy
- Specialized in creating ultra-compact circuit cards to fit within a 1-inch x 2-inch x 0.5-inch enclosure
- Optimized enclosure box for ingress protection, impact resistance, non-flammability, and insulation properties

Programmable Intelligent Collaborative Engagement Munitions (PICEM) | *JHU APL*

May 2023 – August 2023

- Collaborated on a MATLAB-based project optimizing decision-making for non-centralized projectile swarms
- Worked closely with team to refine and integrate object-association and object-assignment algorithms
- Created testing framework to evaluate reliability of PICEM algorithms operating autonomously, tested 50+ edge cases

TECHNICAL SKILLS

- **Platforms/Tools:** Unigraphics/Siemens NX, Ansys, SolidWorks, Ansys Sherlock, MasterCAM, Jira, Confluence, Teamcenter
- **Skills:** Structural Simulation, CNC Machining, DFM/DFA/DFS, GD&T, Plastic/Composite Manufacturing, Injection Molding
- **Programming Languages:** Python, Java, MATLAB, C++, OCaml, R, Git, Arduino